RedstoneLab

Redstone meters and piston previews, exact to the tick



Minecraft 1.21.11 · Fabric · 0.6.1

by **kyoubitt**

1.1 Install

- **Fabric Loader** + **Fabric API** for 1.21.11.
- `redstonelab-x.y.z.jar`, newest release (not `-sources`), into `mods`.
- Nothing else needed. Mod Menu optional.

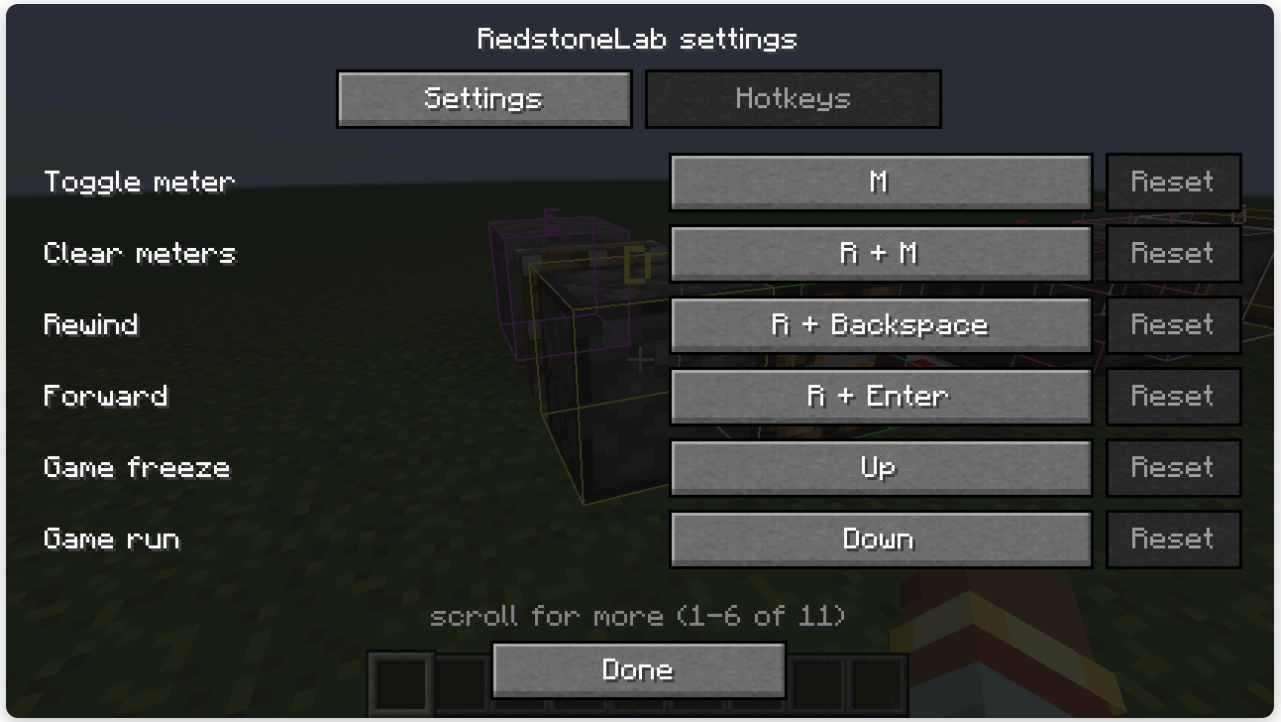
> `github.com/kyoubitt/RedstoneLab/releases`

On	You get
Client only	Piston previews, any server
Client + server	Everything, exact ticks
Server only	Meters as boss bars / chat, piston previews

1.2 Keys

`M`	Meter on / off
`R` + `M`	Remove all
`Up`	Freeze game / step 1t
`Down`	Normal speed
`Left`	Slow motion
`Right`	Faster
`R` + `Backspace`	1 tick back
`R` + `Enter`	1 tick forward
`P`	Hide piston previews
`R` + `G`	Meter groups
`R` + `C`	Settings

> Change: Settings → Hotkeys, click one, hold the keys, let go. `Esc` unbinds.



1.3 Look at redstone



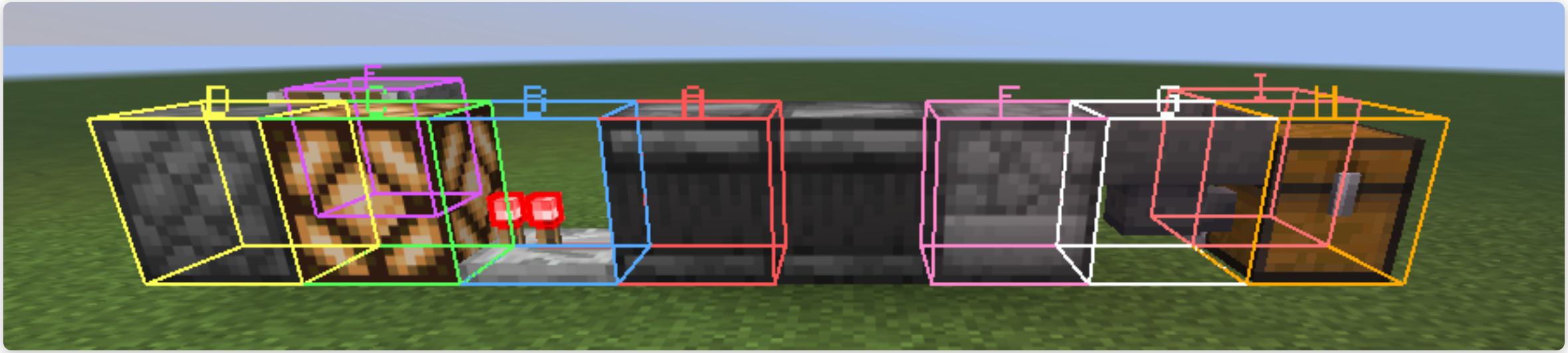
1. What it does **now** and **why**. Here its input just went off. On: `Repeater · on · delay 2t · powered by Observer (west, 15)`
2. The **rule** behind it, in one line.

- Dust is traced back to its source: `power 13 · 2 blocks from Repeater (15)`
- Pistons: `powered from above (QC)`, `can't push: Obsidian can't move`, `BUD`
- Torches: `burnt out`. Blocks: strong or weak power.

> Any server. Off: Settings → **Look at redstone**.

2.1 Meters

- `M` on a block = meter. `M` again removes it. `R` + `M`: remove all.
- Any block: dust, repeaters, pistons, observers, droppers, hoppers, chests…
- Letter + colour per meter. Follows blocks moved by pistons.



2.2 Timeline



1. **Order**, from A: `A → 2t → B D F G` = B, D, F, G 2t after A
2. Meter letter
3. **Bar** = on, number = ticks
4. Block + on / off lengths

`6t` exact **`3t/5t`** alternates **`2t/3t/3t`** pattern **`2t or 3t`** no pattern **`thin line`** powered, not on **`white line`** 0-tick

2.3 Per block

Block	Bar	Text
Dust, lamp, observer	on	`on 2t · off 4t`
Repeater	on	`delay 4t · locked`
Torch	lit	`burnt out` (8 flips in 60t)
Plain block	powered	`strong 15` lights dust · `weak 13` doesn't
Piston	out	`moves every 6t · out 2t`
Dropper, dispenser, crafter	fires	`27 Stone · signal 1 · fires every 4t`
Hopper	locked	`in / out every 8t`
Chest, barrel…	signal > 0	`signal 1 (2 at 124)` = 2 at 124 items. `blocked`: reads 0
Comparator	on	`subtract · back 15 · side 14`
Target, daylight, sculk, plate	on	`power 7`
Composter, cauldron, cake…	powered	`signal 3` = comparator reads 3

2.4 Sub-tick order



1. `R` + `Backspace` / `Enter`: 1 tick back / forward. Back at now it runs again.
2. What happened that tick, **in game order**, by part of the tick.

> `Up` freezes the game (op), again = step 1 tick. `Left` / `Right`: slow motion. `Down`: normal. The timeline shows which keys do what. Block / shape updates: Settings → **Show event marks**.

2.5 Commands

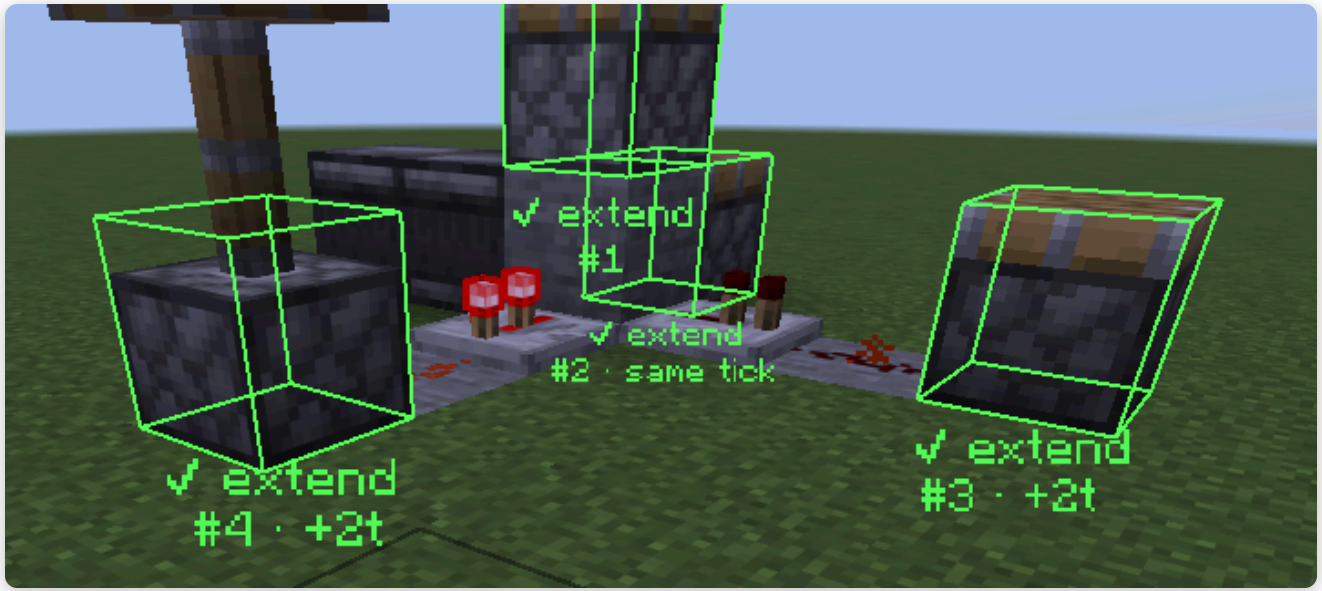
`/rlab meter expect A 2 4`	On 2t, off 4t. Wrong = red.
`/rlab meter name A clock`	Name for A.
`/rlab group <name>`	Saved meter sets. `R` + `G` lists them.
`/rlab meter`	Meters in chat (no client mod needed).
`/rlab tool`	Right-click tool, no keys needed.

3.1 Piston previews



- **Empty hand, right-click a piston.** Again hides it. `P` hides all.
- ✔ **push 3/12**: numbers = move order. `breaks` = destroyed.
- ✖ **blocked**: `stuck` = the block in the way.
- ✖ **too many**: `13th` = the extra block.
- ⚡ **QC** powered from above.
- ⚡ **BUD** moves on next update.
- Any server: only your client needs it.

3.2 Firing order

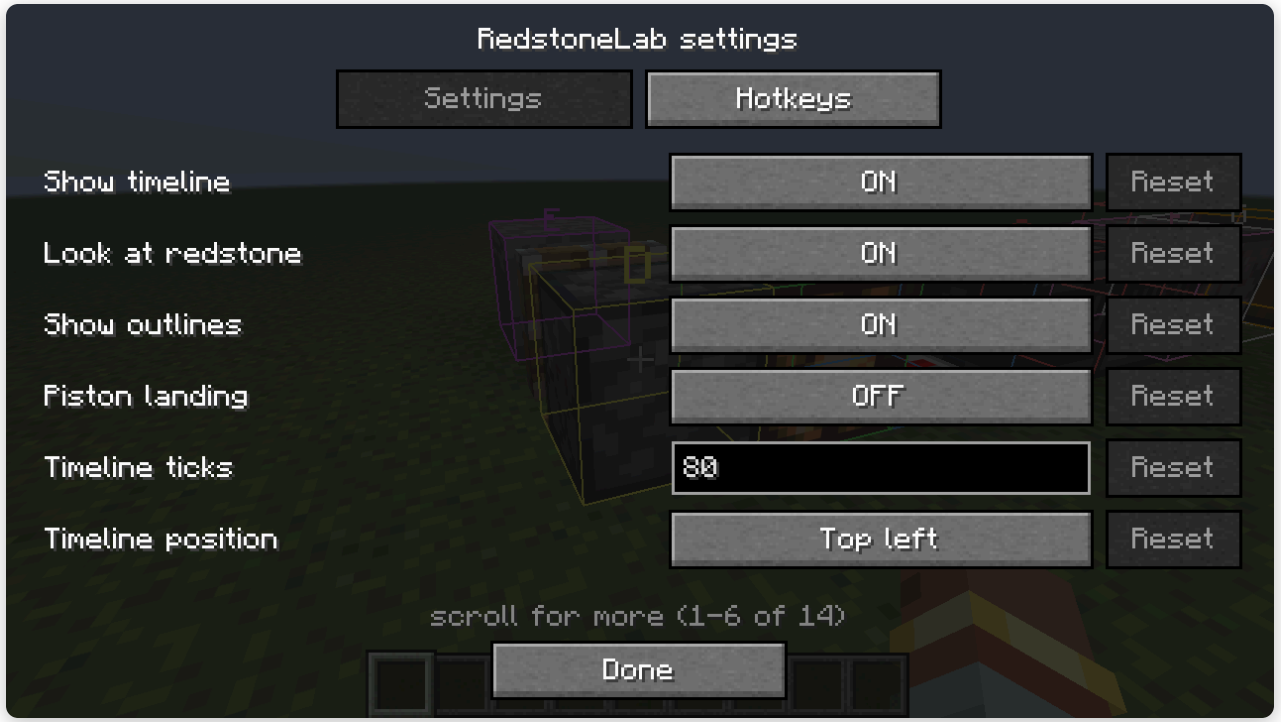


- Several shown: order they extend, from the first one shown.
- `+2t` = 2t after the one before.
- `same tick` = same tick, moved after it.
- `own timing` = different speed.

> Exact with the mod on the server. Otherwise lag can shift ticks by 1.

4 Settings

- `R` + `C` or Mod Menu.
- **Settings**: timeline on / off, look at redstone, outlines, piston landing boxes, ticks shown, where and how big, background, event marks.
- **Hotkeys**: any key or combo. `Esc` unbinds.
- Saves instantly. Hover for help.



Links

Download & source	`github.com/kyoubitt/RedstoneLab`
Help in game	`/rlab help`
License	LGPL-3.0